Tricky stuff

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Contents

1 Everything

1.1 Lebesque-measurable set that is not Borel

Let $\mathcal{C} \subset [0,1]$ be Cantor set, $c\colon [0,1] \to [0,1]$ be cantor function defined by

$$\begin{gathered} c = \frac{1}{2} \text{ on } \left[\frac{1}{3}, \frac{2}{3}\right] \\ c = \frac{1}{4} \text{ on } \left[\frac{1}{9}, \frac{2}{9}\right],\ c = \frac{3}{4} \text{ on } \left[\frac{7}{9}, \frac{8}{9}\right] \\ \vdots \\ c\left(\sum_{n\in\mathbb{N}} \frac{2a_n}{3^n}\right) = \sum_{n\in\mathbb{N}} \frac{a_n}{2^n} \quad \forall \{a_n\}_{n\in\mathbb{N}},\ a_n \in \{0,1\} \\ c(x) = \sup_{y\le x} c(y) \text{ for } x \in [0,1] \setminus \mathcal{C} \end{gathered}$$

Then c is continuous, surjective and nondecreasing. Therefore

$$\psi(x) := c(x) + x \tag{1}$$

is continuous, increasing and bijective $[0,1] \to [0,2]$. Since

$$
\begin{aligned}
[0,1] \setminus \mathcal{C} &= \bigcup_{n \in \mathbb{N}} (a_n, b_n) \text{ where } (a_n, b_n) \text{ are intervals over which } c(x) \text{ is constant} \\
\lambda\left(\psi\left(a_n, b_n\right)\right) &= \lambda(c(a_n) + a_n, c(b_n) + b_n) = \lambda(c|_{A_n} + a_n, c|_{A_n} + b_n) = \lambda(a_n, b_n) \\
\Rightarrow \lambda(\psi([0,1] \setminus \mathcal{C})) &= \lambda([0,1] \setminus \mathcal{C}) = 1 \\
\Rightarrow \lambda(\psi(\mathcal{C})) &= \lambda([0,2] \setminus \psi([0,1] \setminus \mathcal{C})) = 1
\end{aligned}
$$

By Vitali's theorem, any Lebesque-measurable set of positive measure has a non-measurable subset. Denote

$$N \subset \psi(\mathcal{C}) \tag{2}$$

such a non-measurable subset. Then $\psi^{-1}(N) \subset \mathcal{C}$ is Lebesque-measurable since $\lambda(\mathcal{C}) = 0$. But ψ^{-1} is increasing, therefore Borel, so if

$$A := \psi^{-1}(N) \tag{3}$$

was Borel-measurable, also

$$N = (\psi^{-1})^{-1}(A)$$

would be, contradiction. Therefore A is Lebesque-measurable but not Borel-measurable.

On the other hand, the completion of Borel sigma algebra is easily seen to be Lebesque sigma algebra:

Denote $\mathcal{B}$ Borel algebra, $\mathcal{L}$ Lebesque algebra, $\overline{\mathcal{B}}^{\lambda}$ the completion of $\mathcal{B}$ w.r.t. Lebesque measure λ, G_δ to be set of countable intersections of open sets ("G_δ" sets). Clearly $\overline{\mathcal{B}}^{\lambda} \subset \mathcal{L}$. To prove the other inclusion, choose arbitrary $A \in \mathcal{L}$. Then $\exists B \in G_\delta \colon A \subset B,\ \lambda(B \setminus A) = 0$, so

$$A = \underbrace{B}_{\in \mathcal{B}} \setminus \underbrace{(B \setminus A)}_{\in \overline{\mathcal{B}}^{\lambda},\text{ since } \lambda(B \setminus A) = 0} \in \overline{\mathcal{B}}^{\lambda}$$

1.2 Composition of Lebesque measurable functions is not Lebesque measurable

By definition, $f \colon \mathbb{R} \to \mathbb{R}$ is Lebesque measurable if it is measurable as a map $(\mathbb{R}, \mathcal{L}) \to (\mathbb{R}, \mathcal{B})$. Let $\psi \colon [0,1] \to [0,2]$ be as in (1), then since $[0,1]$ is compact ψ^{-1} is also continuous (therefore Lebesque measurable). Denote $A \subset [0,1]$ a non-Borel set from (3), then since $\lambda(A) = 0$, $\mathbb{1}_A \colon [0,1] \to \mathbb{R}$ is Lebesque measurable. But

$$(\mathbb{1}_A \circ \psi^{-1})^{-1}(\{1\}) = (\psi^{-1})^{-1}(A) = N$$

is the non-measurable set from (2), therefore $\mathbb{1}_A \circ \psi^{-1}$ is not measurable.

Meaning: this is why we define Carathéodory functions.

1.3 Hölder continuous function not absolutely continuous

Let c be the cantor function and c_k be defined by

$$
\begin{aligned}
& c_k\left(\sum_{n=1}^{k} \frac{2a_n}{3^n}\right) = \sum_{n=1}^{k} \frac{a_n}{2^n} \quad \forall \{a_n\}_{n=0}^{k}, a_n \in \{0,1\} \\
& c_k \text{ is piecewise linear on } [0,1] \\
& c_k(0) = 0, c_k(1) = 1
\end{aligned}
$$

Then

- $\|c - c_k\|_\infty = 2^{-k}$
- c_k is Lipschitz with constant $\left(\frac{3}{2}\right)^k$ (think about the slope of steepest line)

So for $x, y \in [0,1]$

$$\begin{aligned} |c(x) - c(x)| &\leq \|c - c_k\|_\infty + |c_k(x) - c_k(y)| + \|c - c_k\|_\infty \\ &\leq 2 \cdot 2^{-k} + \left(\frac{3}{2}\right)^k |x - y| \end{aligned}$$

Choose $k \in \mathbb{N}$ such that $3^{-k-1} \leq |x - y| < 3^{-k}$, then

$$|c(x) - c(y)| \leq 3 \cdot 2^{-k} = 6 \cdot 2^{-k-1} = 6 \cdot 3^{-(k-1)\cdot \log_3 2} < 6 \cdot |x - y|^{\log_3 2}$$

So the cantor function is $\log_3 2$-Hölder continuous.
Let $\epsilon > 0$. Since c is locally constant on $[0,1] \setminus \mathcal{C}$, which is a countable union of open intervals with measure one, by continuity of measure we can choose $x_1, ..., x_n, y_1, ..., y_n \in [0,1], x_k < y_k$ (in fact they will be endpoints of form $\frac{l}{3^n}, l, n \in \mathbb{N}$) such that

$$\sum_{k=1}^{n} |y_k - x_k| > 1 - \epsilon \quad \text{and c is constant on intervals} \quad (x_k, y_k)$$

Since c is constant on big set, and $c(0) = 0, c(1) = 1$, it must grow fast on a small set:

$$|x_1 - 0| + \sum_{k=1}^{n-1} |x_{k+1} - y_k| + |1 - y_n| = 1 - \sum_{k=1}^{n} |y_k - x_k| < \epsilon$$

$$\text{BUT} \quad c(x_1) - c(0) + \sum_{k=1}^{n-1}(c(x_{k+1}) - c(y_k)) + c(1) - c(y_n) = 1 - \sum_{k=1}^{n}(c(y_k) - c(x_k)) = 1$$

so c is not $AC[0,1]$.

A shorter proof might use characterization $AC[0,1] = W^{1,1}(0,1)$, since c has classical derivative zero almost everywhere, but certainly

$$c(0) + \int_0^1 c'(x)dx = 0 + \int_0^1 0 \; dx = 0 \neq c(1)$$

In fact, $c \in BV[0,1] \setminus AC[0,1]$ and it's Lebesque-Stieltjes measure is singular nonatomic with respect to the Lebesque measure on $[0,1]$.

1.4 Hopf fibration, dirac belt

$$S^1 \to S^3 \cong B_{1,\mathbb{C}^2}(0) \to S^2 \cong \mathbb{PC}^1$$

Hilbert space for two-state system is

$$\mathcal{H} = \mathbb{C}^2 = \mathbb{C} \left|\uparrow\right\rangle \oplus \mathbb{C} \left|\downarrow\right\rangle$$

When we mod out the norm, we are left with $B_{1,\mathbb{C}^2}(0) \cong S^3$ and when we mod out the phase, we are left with $\frac{S^3}{S^1} \cong S^2$. Quantum evolution happens on S^3 and state space is $S^2 = \mathbb{P}(\mathcal{H})$.

Below we use pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \vec{\sigma} = \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix}$$

The Hopf fibration is the map which sends spinors (elements of S^3) to the expectation values of spin in the x, y, z directions:

$$\langle \bullet | \vec{\sigma} | \bullet \rangle \colon S^3 \to S^2$$

$$\alpha \left|\uparrow\right\rangle + \beta \left|\downarrow\right\rangle = \left|\psi\right\rangle \mapsto \left\langle\psi\right| \vec{\sigma} \left|\psi\right\rangle = \begin{pmatrix} \langle\sigma_x\rangle_\psi \\ \langle\sigma_y\rangle_\psi \\ \langle\sigma_x\rangle_\psi \end{pmatrix}$$

$$\text{or:} \quad \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \mapsto \begin{pmatrix} \alpha\bar{\beta} + \beta\bar{\alpha} \\ i(\alpha\bar{\beta} - \beta\bar{\alpha}) \\ |\alpha|^2 - |\beta|^2 \end{pmatrix}$$

Generally, given a smooth manifold M, a principal $U(1)$-bundle is a fiber bundle $p\colon E \to M$ with a group action $U(1) \times E \to E$ such that $\forall e \in E, g \in G: \; e \in p^{-1}(x) \Rightarrow g \cdot e \in p^{-1}(x)$ and the action on each fiber is free and transitive. In our case $M = S^2$.

1.5 Examples of adjunctions

$Unif$ is category of uniform spaces and uniformly cts functions, $CUnif_{T2}$ is category of complete hausdorff uniform spaces, $\hat{A}$ is completion of A.

$$Hom_{Unif}(A, Y) \cong Hom_{CUnif_{T2}}(\hat{A}, Y)$$

1.6 Homeomorphism doesn't preserve completeness

Completeness is not a topological invariant. The spaces $(0, 1)$, $\mathbb{R}$ with Euclidean topology are homeomorphic via

$$h\colon \mathbb{R} \to (0,1), \quad h(x) = \frac{\tanh(x) + 1}{2}$$

$\mathbb{R}$ is complete, $(0, 1)$ is not.

1.7 Fréchet space that is not normable

Consider $L^1_{loc}(\mathbb{R}^d)$ given the Fréchet topology generated by seminorms

$$|f|_n = \int_{\|x\| \le n} |f(y)| dy$$

Suppose there is a norm $\|\bullet\|$ generating the same topology. Then $\exists C > 0, C_n > 0, n \in \mathbb{N}$ such that

$$\|\bullet\| \le C |\bullet|_N, \tag{4}$$
$$|\bullet|_n \le C_n \|\bullet\| \tag{5}$$

from this we get $|\bullet|_{N+1} \le C_{N+1} C |\bullet|_N$, which is clearly wrong for the function $f = \chi_{\{N < \|x\| < N+1\}}$.

To prove (4), for contradiction, assume

$$\forall n, k \in \mathbb{N} \; \exists \phi_{n,k} \in L^1_{loc} : \|\phi_{n,k}\| > n |\phi_{n,k}|_k$$

define $\psi_n = \dfrac{\phi_{n,n}}{n|\phi_{n,n}|_n}$, then

$$\forall l \in \mathbb{N}, n \ge l : |\psi_n|_l \le |\psi_n|_n = \frac{1}{n} \to 0 \qquad n \to \infty$$

so $\psi_n \to 0$ in L^1_{loc} but $\|\psi_n\| > 1$, contradiction. Generally, for any Fréchet space X where the seminorms are nondecreasing, the inequalities (1) and (2) hold, so if we can find elements $x \in X$ such that $|x|_{N+1}$ can be arbitrarily large and $|x|_N$ arbitrarily small, the space is nor normable.

For $C_0^\infty[0, 1]$ where topology is generated by $|f|_n = \sup_{k \le n} \|f^{(k)}\|_\infty$, the element $f(x) = \frac{1}{n^N} \sin(nx)$ (n large) has small $|\bullet|_{N-1}$ but large $|\bullet|_N$.

1.8 Banach space with no preduals

Suppose $X^* = L^1(0,1)$. Then by Banach-Alaoglu, $B_X = \{f \in L^1(0,1)\colon \|f\| \leq 1\}$ is w^*-compact and convex, so by Krein-Milman $B_X = \overline{\text{aconv}}\ \text{ext}\ B_X$. But there are no extreme points of B_X :

Suppose $f \in B_X$ is extreme. Since $\int_0^x |f(t)|dt$ is continuous in x,

$$\exists s \in (0,1) : \int_0^s |f(t)|dt = \frac{\|f\|_1}{2}$$

Now $g_1 := 2f\chi_{(0,s)},\ g_2 := 2f\chi_{(s,1)}$ satisfy $f = \dfrac{g_1+g_2}{2},\ g_1, g_2 \in B_X,\ g_1, g_2 \neq f.$

1.9 Weak* convergence $\not\Rightarrow$ Weak convergence

Let $X = c_0,\ X^* = l^1, X^{**} = l^\infty$. Then the Shauder basis

$$e_n = (0,0,...,0,\underbrace{1}_{n\text{'th position}},0,...)$$

satisfies

$$\begin{aligned}&\forall x \in c_0, \langle e_n, x\rangle \to 0\\ &\text{For } y = ((-1)^n)_{n=1}^\infty \in l^\infty, \langle y, e_n\rangle = (-1)^n \not\to 0\end{aligned}$$

So $e_n \overset{w^*}{\to} 0$ but $e_n \overset{w}{\not\to} 0$.

1.10 Weakly continuous map not continuous

By Hahn-Mazurkiewicz theorem, a T_2 space X is a continuous image of $[0,1]$ iff X is compact, connected, locally connected, and second-countable (equivalently: compact, connected, locally connected, metrizable.) Taking X to be a unit ball in an infinite dimensional Banach space, we see that there exists a map $f\colon [0,1] \to X$ such that

$$\forall t_0 \in [0,1]\colon f(t) \overset{w}{\to} f(t_0) \text{ as } t \to t_0, \quad \text{BUT} \quad f(t) \not\to f(t_0) \text{ as } t \to t_0.$$

1.11 Almost everywhere convergence is not topological

It holds in any topological space (X,τ) and sequence $\{x_n\}_{n\in\mathbb{N}} \subset X$ it holds

$$x_n \overset{\tau}{\to} x \iff \begin{array}{l}\text{For any subsequence } \{x_{n_k}\}_{k\in\mathbb{N}} \text{ of } \{x_n\},\\ \text{there is a subsequence } \{x_{n_{k_l}}\}_{l\in\mathbb{N}} \text{ such that } x_{n_{k_l}} \overset{\tau}{\to} x.\end{array}$$

Consider the Haar system

$$\begin{aligned}\psi_{n,k} &:= \mathbb{1}_{\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right)}, \quad 0 \leq k \leq 2^n - 1, \quad n \in \mathbb{N}.\\ (\phi_l)_{l\in\mathbb{N}} &:= (\psi_{1,1},\ \psi_{2,1},\ \psi_{1,2},\ \psi_{2,2},\ \psi_{3,2},\ \psi_{4,2},\ \psi_{1,3},\ \ldots)\end{aligned}$$

Then $\phi \overset{a.e.}{\not\to} 0$, but for any subsequence of $\phi_n = \mathbb{1}_{I_n}$ we can choose a subsequence $\phi_{n_k} = \mathbb{1}_{I_{n_k}}$ such that $\sum_{k\in\mathbb{N}} |I_{n_k}| < \infty$, and therefore from Borel-Cantelli

$$\begin{aligned}&\lambda_1(\limsup_{k\to\infty} I_{n_k}) = 0\\ \Rightarrow\ &\lambda_1(\{x \in [0,1]\colon x \text{ lies only in finitely many of } I_{n_k}\}) = 1\\ \Rightarrow\ &\phi_{n_k} \overset{a.e.}{\to} 0\end{aligned}$$

1.12 Symmetric operator with non-real spectrum

Let $A\colon L^2(0,1) \to L^2(0,1)$ be

$$Au := u'', \quad D(A) := \{u \in W^{2,2}(0,1)\colon u(0) = u'(0) = u(1) = u'(1) = 0\}.$$

Clearly A is symmetric. It's adjoint is

$$A^*u = u'', \quad D(A^*) = W^{2,2}(0,1)$$

For any $\lambda \in \mathbb{C}$, equation $A^*u_\lambda = \lambda u_\lambda$ has a two-dimensional space of solutions

$$u_\lambda(x) = C_1 e^{\sqrt{\lambda}x} + C_2 e^{-\sqrt{\lambda}x}$$

And therefore

$$(\operatorname{Im}(\lambda\mathbb{1} - A))^\perp = \operatorname{Ker}(\overline{\lambda}\mathbb{1} - A^*) \neq 0 \quad \Rightarrow \sigma(A) = \mathbb{C}$$

1.13 Momentum operator on half-line does not exist

Let $A\colon L^2(0,\infty) \to L^2(0,\infty)$ be

$$Au := iu', \quad D(A) := \{u \in W^{1,2}(0,\infty)\colon u(0) = 0\}$$

To show A is symmetric, in the following computation

$$\forall u, \varphi \in D(A)\colon (Au, \varphi) = \int_0^\infty iu'\overline{\varphi}\, dx = \int_0^\infty u\overline{i\varphi'}\, dx + i[u\overline{\varphi}]_0^\infty = (u, A\varphi) + i(u\overline{\varphi})|_\infty$$

we need that $\lim\limits_{x\to\infty} u(x)\overline{\varphi(x)} = 0$. We show that for all $v \in W^{1,2}(0,\infty)$, $\lim\limits_{x\to\infty} v(x) = 0$.

$$\forall y < x \in \mathbb{R}\colon v^2(x) - v^2(y) = \int_y^x 2v(t)v'(t)dt \leq 2\|v\|_{L^2(y,\infty)}\|v'\|_{L^2(0,\infty)} \overset{y\to\infty}{\to} 0$$

therefore $v(y)$ is Cauchy for $y \to \infty$, and so $\lim\limits_{y\to\infty} v(y)$ exists. From $v \in L^2(0,\infty)$ it follows $\lim\limits_{y\to\infty} v(y) = 0$. Therefore A is symmetric.

To calculate the defect indices

$$\begin{aligned} n_+(A) &:= \dim \operatorname{Im}(A + i\mathbb{1})^\perp = \dim \operatorname{Ker}(A^* - i\mathbb{1}), \\ n_-(A) &:= \dim \operatorname{Im}(A - i\mathbb{1})^\perp = \dim \operatorname{Ker}(A^* + i\mathbb{1}) \end{aligned}$$

we need to calculate A^*:

$$\begin{aligned} D(A^*) &= \{\varphi \in L^2(0,\infty)\colon \exists \psi \in L^2(0,\infty)\ \forall u \in D(A)\colon (Au, \varphi) = (u, \psi)\} \\ \forall u \in D(A)\colon (Au, \varphi) &= \int_0^\infty iu'\overline{\varphi}\, dx = \underbrace{i[u\overline{\varphi}]_0^\infty}_{=0} + \int_0^\infty u\overline{i\varphi'}dx \ \Rightarrow\ D(A^*) = W^{1,2}(0,\infty) \\ n_+(A) &= \dim\{u \in W^{1,2}(0,\infty)\colon iu' - iu = 0\} = \dim(\operatorname{span}\{e^x\} \cap W^{1,2}(0,\infty)) = 0 \\ n_-(A) &= \dim\{u \in W^{1,2}(0,\infty)\colon iu' + iu = 0\} = (\operatorname{span}\{e^{-x}\} \cap W^{1,2}(0,\infty)) = 1 \end{aligned}$$

Since $n_+ \neq n_-$, there is no self-adjoint extension.

1.14 Fourier series of continuous function does not converge

. Let for $f\colon [0,2\pi] \to \mathbb{R}$

$$S_N f(x) = \sum_{n=-N}^{N} a_n e^{-inx} = \frac{1}{2\pi}\int_0^{2\pi} D_N(x-t) f(t)dt,$$

where D_N is the Dirichlet kernel:

$$D_N(x) = \frac{\sin\left(\frac{2N+1}{2}x\right)}{\sin\left(\frac{x}{2}\right)}.$$

It is well-known that

- for $f \in L^p(0,2\pi)$, $S_N f \to f$ in $L^p(0,2\pi), 1 < p < \infty$,
- for $f \in BV[0,1]$, $S_N f \to f$ pointwise on $[0,2\pi]$,
- for $f \in AC[0,1]$, $S_N f \rightrightarrows f$ on $[0,2\pi]$.

Fix $x \in [0,2\pi], N \in \mathbb{N}$. Then define a signed Radon measure

$$\Phi_{x,N}\colon C[0,2\pi] \to \mathbb{R}, \quad \Phi_{x,N} f := S_N f(x)$$

it's norm is

$$||\Phi_{x,N}||_{(C[0,2\pi])^*} = \frac{1}{2\pi}\int_0^{2\pi} |D_N(s)|dt$$

But

$$\frac{1}{2\pi}\int_0^{2\pi} |D_N(s)|dt \geq \frac{1}{2\pi}\int_0^{2\pi} \frac{|\sin\left(\frac{2N+1}{2}s\right)|}{\frac{s}{2}}ds = \int_0^{(2N+1)\pi} \frac{|\sin x|}{x}dx \overset{N\to\infty}{\to} \int_0^{\infty} \frac{|\sin x|}{x}dx = \infty$$

So $\{\Phi_{x,N}\}_{N\in\mathbb{N}}$ is not bounded in $(C[0,2\pi])^*$. By Uniform Boundedness Principle we get

$$\forall x \in [0,2\pi]\ \exists \text{ dense set } \mathcal{A} \subset C[0,2\pi] \text{ such that } \forall f \in \mathcal{A}\colon |S_N f(x)| \overset{N\to\infty}{\to} \infty \tag{6}$$

1.15 Functional not attaining it's norm. Counterexample to projection theorem

Let $X = L^1(0,1)$, $\phi \in X^*$ defined by $\langle \phi, f\rangle := \int_0^1 xf(x)dx$. We will show

$$\nexists f \in X\colon \|f\|_X = 1,\ |\langle \phi, f\rangle| = \|\phi\|_{X^*}.$$

Clearly

$$\|\phi\|_{(L^1(0,1))^*} = \|x\|_{L^\infty(0,1)} = 1.$$

For contradiction, suppose there exist $f \in L^1(0,1)$ such that $\|f\|_1 = 1$, $|\langle \phi, f\rangle| = 1$. Then f does not change sign on $(0,1)$ so we can choose f such that $f \geq 0$ on $(0,1)$ (if not, then $\||f|\|_{L^1(0,1)} = \|f\|_{L^1(0,1)} = 1$, $|\langle \phi, |f|\rangle| > |\langle \phi, f\rangle| = 1$, contradicting $\|\phi\|_{(L^1)^*} = 1$). Then

$$\begin{aligned}
1 &= \int_0^{1-\frac{1}{n}} xf(x)\,dx + \int_{1-\frac{1}{n}}^1 xf(x)\,dx \leq \\
&\leq \left(1-\frac{1}{n}\right)\int_0^{1-\frac{1}{n}} f(x)\,dx + \int_{1-\frac{1}{n}}^1 f(x)\,dx \\
&= \left(1-\frac{1}{n}\right)\underbrace{\int_0^1 f(x)\,dx}_{=1} + \frac{1}{n}\int_{1-\frac{1}{n}}^1 f(x)\,dx
\end{aligned}$$

$$\Rightarrow \quad \int_{1-\frac{1}{n}}^{1} f(x)\,dx \geq 1 \qquad \forall n \in \mathbb{N}$$

$$\downarrow n \to \infty \quad (f \in L^1)$$

$$0 \geq 1.$$

Intuitively, any f attaining the bound $|\langle \phi, f \rangle| = 1$ would have to have all it's mass at $x = 1$, since if the mass is anywhere else, we can move it closer to $x = 1$ to make $\langle \phi, f \rangle$ bigger. This is impossible on the level of L^1 functions. However, taking $\delta_1 \in (C[0,1])^*$, $\langle \delta_1, \varphi \rangle = \varphi(1)$, by Hahn-Banach

$$\|\delta_1\|_{(C[0,1])^*} = 1,\ C[0,1] < L^\infty(0,1) \Rightarrow \exists T \in (L^\infty(0,1))^* \colon T|_{C[0,1]} = \delta_1,\ \|T\|_{(L^\infty(0,1))^*} = 1.$$

And (equating $\phi \in (L^1)^*$ with $x \in L^\infty$):

$$\langle T, \phi \rangle = \langle T, x \rangle = \langle \delta_1, x \rangle = 1.$$

Therefore ϕ attains it's maximum on the unit ball of dual $(L^\infty)^*$, but not on predual L^1. The problem is lack of reflexivity. By James' theorem, for any Banach space Y

$$Y \text{ is reflexive } \iff \text{ any } \phi \in Y^* \text{ attains it's norm on the unit ball } B_1(0) \subset Y.$$

We say that a subspace $A \subset Y$ of Banach space is *proximital*, if

$$\forall x \in Y\ \exists! a \in A \colon \|x - a\| = \inf_{b \in A} \|x - b\| \quad \ldots a := P_A(x)$$

by the Projection theorem, every nonempty closed convex subset of a Hilbert space is proximital. We show that in $X = L^1$, $V := \{f \in L^1(0,1) \colon \langle \phi, f \rangle = 1\}$ is not proximital, even though it is a closed convex subset. We wish to project $0 \in L^1$ to V. Clearly

$$\forall \epsilon > 0\ \exists f_\epsilon \in L^1 \colon \|f_\epsilon\|_{L^1} \leq 1 + \epsilon,\ \langle \phi, f_\epsilon \rangle = 1 \quad \& \quad \forall f \in L^1 \colon \langle \phi, f \rangle = 1 \Rightarrow \|f\|_{L^1} \geq 1$$

$$\Rightarrow \inf_{f \in V} \|0 - f\| = \inf_{f \in V} \|f\| = 1$$

But we know that this bound cannot be attained, so the projection does not exist:

$$f = P_V(0) \ \Rightarrow \ \|f\|_{L^1} = 1 \ \ \& \ \ f \in V, \quad \text{a contradiction.}$$

1.16 $\int_A \int_B f(x,y)dxdy \neq \int_B \int_A f(x,y)dydx$

Consider

$$f(x,y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$